

Evaluating Qwen3 Thinking Capabilities Outside the Thinking Mode

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Abstract

We investigate whether Qwen3 models, trained to reason during post-training with the special `<think>` token, can exhibit similar reasoning behavior when prompted with natural language cues like "Wait". While `<think>` is expected to perform better, we aim to quantify how well Qwen3 models reason using regular language cues under different thinking budgets.

1 Introduction

Language models often benefit from being guided to reason via intermediate steps. Chain-of-Thought (CoT) prompting has demonstrated that providing models with examples of step-by-step reasoning can significantly enhance their performance on complex tasks (?).

Recent work by (?) investigates the impact of reasoning duration on model performance. They fine-tuned the Qwen2.5-32B-Instruct model on a small dataset of 1000 reasoning examples and employed a budget forcing technique. This approach involves appending the word "Wait" multiple times during inference to encourage the model to continue reasoning. After the number of reasoning tokens reaches the limit, the model is forced to produce a final answer. Their results showed that this approach can improve performance on math benchmarks.

The Qwen3 models (?) introduce a thinking mode between `<think>` and `</think>` tokens. These tokens were integrated into the model during post-training, allowing the model to perform complex, multi-step reasoning tasks more effectively. The technical report demonstrates that using this mode significantly improves performance on reasoning tasks, which is even more strengthened by increasing the thinking budget (the number of allowed tokens inside `<think>` block).

A natural question arises: can Qwen3, which was specifically trained to reason within `<think>`

blocks, also exhibit similar level of reasoning capabilities when prompted with natural language cues like "Wait"? It remains unclear whether (and to what extent) this behavior can be elicited outside thinking blocks.

2 Method

We will conduct experiments with the following setup:

- **Models:** We will use Qwen3-1.7B.
- **Datasets:** We will use SVAMP (?).
- **Configurations:** We will compare three configurations:
 - Qwen3 without thinking mode,
 - Qwen3 without thinking mode and with reasoning forced by "Wait" tokens,
 - Qwen3 with thinking mode.
- **Thinking budget:** Multiple values of thinking budget will be tested.

3 Expected results

We expect the configuration with thinking mode to yield the highest accuracy. As the thinking budget increases, the accuracy should also increase.

We anticipate that the "Wait" token will provide some improvements over the baseline without any thinking process involved, and that the results will also improve as the thinking budget increases.